Beyond the Hairball: The Reddit Community Atlas

What really connects Reddit's communities? An interactive, statistically validated map of shared audiences across 170 subreddits (Oct 2017 – Sep 2018)

CSE3729 Computational Social Media Analysis · BML Munjal University · Academic year 2026–27

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Every number in this report is generated from the project's build artifacts by `tools/build_docs.py`; Appendix A maps each headline figure to the file it comes from.

Abstract

Online communities are often mapped through the audiences they share, but raw shared-audience networks mostly measure size: one person who comments in many places links all of them, and the biggest communities look close to everything. We collected 223,218,978 public Reddit comments from a fixed, rule-based frame of 170 subreddits over 12 months (Oct 2017 – Sep 2018), cleaned them to 210,325,806 comments by 2,446,462 pseudonymised commenters, and built a user × subreddit network. The conventional cosine projection links 99.99% of all subreddit pairs. Comparing each pair's shared commenters with the number expected from community size and user activity alone (the Bipartite Configuration Model, BiCM) and keeping only pairs that exceed it at a 1% false discovery rate leaves 13.9% of pairs. That network has a clear, reproducible structure: 9 communities (weighted modularity 0.577, seed agreement ARI 0.995), stable connector roles, and 82%–85% of links recurring between consecutive quarters. We built the results into an interactive web application, the Community Atlas, with four guided stories: what popularity hides, unexpected neighbours, hidden connectors, and how community ties respond when crowds arrive and leave. The project started as a test of whether cross-community "bridge positions" outlast their occupants. It evolved after the first, cosine-based pipeline proved too dense to measure bridges at all. The findings are descriptive and specific to this sample.

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1. Introduction and research questions

Reddit is organised into hundreds of thousands of topical communities. Which of them share an audience, and why, is a basic question about how online attention is organised. It matters for information spread, community moderation and for any study that treats subreddits as separate populations. The usual way to answer it is a bipartite network of users and communities, *projected* onto the communities: two subreddits are linked when people comment in both.

The project began with a narrower, longitudinal question (the **submitted proposal**): *do cross-community bridge positions persist across quarters even when the people occupying them change?* Our first full pipeline (V1, §7) built that analysis on a cosine-similarity projection. On real data, the projection connected 99.85% of subreddit pairs. It produced only four communities and so only six possible bridge positions, all saturated. The bridge question could not be answered with that measurement lens. The final project therefore asks three questions that the data can support, and keeps the bridge idea as a measured connector role:

- **RQ1 (representation).** How much of the audience overlap between subreddits is explained by community size and user activity alone?
- **RQ2 (structure).** Which communities share *surprisingly* large audiences, and what communities and connectors do those relationships form?
- **RQ3 (change).** How stable are these relationships over time, and how do they respond when a crowd suddenly arrives or leaves?

The deliverable is an end-to-end, reproducible pipeline and an interactive application (the **Community Atlas**) through which an evaluator can explore the evidence.

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2. Literature review

Projections are dense by construction. A one-mode projection of a bipartite network is a union of cliques: every user active in *k* communities adds a link between all *k* of them, and high-degree nodes share neighbours by chance (Saracco et al., 2017; Larremore et al., 2014). The same mechanism explains our 99.9%-dense V1 network. Olson and Neal (2015) met it on Reddit itself and extracted a subreddit "backbone" with the disparity filter of Serrano et al. (2009). That filter compares each edge with a per-node uniform null and ignores the bipartite origin of the weights. Coscia and Rossi (2019, preprint) show that the choice of projection and backbone changes substantive conclusions.

Validating projections against a null model. The Bipartite Configuration Model (Saracco et al., 2015) is a maximum-entropy random network that preserves, on average, every node's degree on both sides: here, each user's number of subreddits and each subreddit's number of commenters. Under it, the number of users shared by two communities follows a Poisson-binomial distribution, so each pair can be tested and the tests corrected for multiplicity with the Benjamini–Hochberg false discovery rate (Saracco et al., 2017). Neal, Domagalski and Sagan (2021) compare alternatives to the exact fixed-degree model and recommend this "stochastic degree sequence" approach with BiCM probabilities when exact sampling is impractical, as it is for 2,446,462 users. Cimini et al. (2022) show that different null models disagree at a fixed p-value but agree at matched density. This is why we report densities and run a sensitivity check rather than trusting one threshold. Applications to social media include validated projections of Twitter user networks (Becatti et al., 2019; Caldarelli et al., 2020). Waller and Anderson (2021) make the same conceptual move for Reddit at scale: communities are related when their user bases are "surprisingly similar" beyond chance.

Communities, roles and time. We detect communities with Leiden (Traag, Waltman & van Eck, 2019), which guarantees connected communities. Good, de Montjoye and Clauset (2010) show that modularity

has many near-optimal partitions, so we report seed agreement and sensitivity rather than a single partition. A subreddit's connector role is measured with the participation coefficient of Guimerà and Amaral (2005). Communities and links are compared across time by Jaccard matching (Greene, Doyle & Cunningham, 2010). Text-free event studies of Reddit measure active users and newcomers around documented dates and mask archive outages (Horta Ribeiro et al., 2021). Crypto-community activity co-moves with prices during bubbles, without implying causation (Phillips & Gorse, 2018).

Visualisation and storytelling. The Atlas follows Shneiderman's (1996) "overview first, zoom and filter, details on demand". Its stories use the "martini glass" structure of Segel and Heer (2010): a short author-driven sequence that opens into free exploration. Node-link diagrams of dense graphs are hard to read (Ghoniem, Fekete & Castagliola, 2005), which is one more reason to show the validated network. Following Archambault, Purchase and Pinaud (2011), time comparisons are shown side by side by default. The force-directed layout is treated as an approximation in which distance is not literal (Jacomy et al., 2014).

Data quality and ethics. Pushshift-derived archives, which the Arctic Shift data repackages (Baumgartner et al., 2020), have bursty missing data; network analyses of such data are among the most exposed to it (Gaffney & Matias, 2018). Social-media samples are not populations, and platform design shapes what can be measured (Ruths & Pfeffer, 2014). Reddit research ethics stresses aggregation and care with sensitive communities (Proferes et al., 2021).

Evidence matrix (selected sources).

Source	Contribution	Decision it informs
Saracco et al. (2015, 2017)	BiCM; Poisson-binomial test of shared neighbours with FDR	The edge definition
Neal et al. (2021); Cimini et al. (2022)	Comparison of null models; agreement at matched density	BiCM over exact sampling; report density and sensitivity
Larremore et al. (2014)	Projections lose information; bipartite SBM	Rejected direct SBM on 2,446,462 users as disproportionate
Serrano et al. (2009); Olson & Neal (2015)	Disparity-filter backbones, including on Reddit	V1 display baseline; Reddit precedent
Waller & Anderson (2021)	Reddit communities related by beyond-chance overlap	Supports observed ÷ expected
Traag et al. (2019); Good et al. (2010)	Leiden; degeneracy of modularity	Leiden, seed and sensitivity reporting
Guimerà & Amaral (2005)	Participation coefficient	Connector score
Greene et al. (2010)	Jaccard matching across snapshots	Link and community persistence
Shneiderman (1996); Segel & Heer (2010)	Overview → details; martini-glass stories	App structure and story mode
Archambault et al. (2011); Jacomy et al. (2014); Ghoniem et al. (2005)	Small multiples; layout caveats; dense-graph readability	Side-by-side periods; layout caption; no hairball as primary map
Gaffney & Matias (2018); Baumgartner et al. (2020)	Missing data in Pushshift-derived corpora; provenance	Counts as lower bounds; provenance statement
Horta Ribeiro et al. (2021); Phillips & Gorse (2018)	Text-free event metrics; crypto activity co-moves with price	Story D metrics and wording
Ruths & Pfeffer (2014); Proferes et al. (2021)	Population bias; Reddit research ethics	Scope wording; aggregate-only publication

3. Data

Source. Public Reddit comments from the Arctic Shift archive, a re-packaging of Pushshift dumps hosted on Hugging Face (`open-index/arctic`), read remotely with DuckDB. Comments in this archive end in August 2022.

Sampling frame. 170 subreddits chosen by a fixed rule before any analysis: 25 in each of six topical categories (politics & news, hobby, support, finance, expertise, media) plus 20 front-page "gateway" hubs, each with at least 2,000 comments in every one of three calibration months (`frame.csv`, `FRAME_JUSTIFICATION.md`).

Scope versus proposal. The proposal planned roughly 150 subreddits over 2019–2025; verification showed comments are only available up to 2022-08, and the window was revised to 2018–2022. This bounded first study processed **12 months, Oct 2017 – Sep 2018 (2017 Q4, 2018 Q1, 2018 Q2, 2018 Q3)**.

2017 Q4 was added to the original three 2018 quarters so the window contains the December 2017 Bitcoin peak. The later years were not collected. All findings describe this sample and window, not Reddit as a whole.

Collection and cleaning. Usernames are replaced in the query by a salted SHA-256 hash; no text is kept, only author hash, subreddit, timestamp and a moderator flag. Every fetch asserts that all 170 subreddits are present (coverage: 2017 Q4 complete, 2018 Q1 complete, 2018 Q2 complete, 2018 Q3 complete). Cleaning keeps accounts with at least five comments in a quarter and removes accounts that are both very high-volume (≥ 2,000 comments) and active in at least half the frame (bots).

Quarter	Raw comments	Retained	% kept	Commenters (raw → kept)	Bot accounts removed
2017 Q4	51,616,108	48,633,963	94.2%	2,763,494 → 1,144,505	5
2018 Q1	55,248,128	52,118,250	94.3%	2,919,421 → 1,215,022	5
2018 Q2	57,301,138	54,002,756	94.2%	3,069,238 → 1,279,269	5
2018 Q3	59,053,604	55,570,837	94.1%	3,248,860 → 1,339,394	0
Total	**223,218,978**	**210,325,806**	**94.22%**	2,446,462 distinct kept	15

4. Methods

1. **Bipartite matrix.** For any period (the whole window, a quarter or a month), a binary matrix records which users commented in which subreddits. Duplicate records are collapsed; this matters, because unsummed duplicates inflate user degrees and break the model.
2. **Raw overlap (baseline).** Cosine similarity of commenter sets, as in V1.
3. **Expected overlap.** The BiCM probability $p(u, s)$ that user u comments in subreddit s, fitted so every user keeps their number of subreddits and every subreddit its number of commenters (on average). Users with the same degree share parameters, so the fit has at most a few hundred unknowns. Expected shared commenters: $E(a, b) = \Sigma_u\, p(u, a) \cdot p(u, b)$. **Affinity** = observed shared commenters ÷ $E(a, b)$.
4. **Validated ("surprising") links.** One-sided Poisson test of the observed count against $E(a, b)$, the reference implementation's default approximation, with Benjamini–Hochberg FDR at 1% over all 14,365 pairs.
5. **Communities and roles.** Leiden (RBConfiguration, $\gamma = 1$) on log-affinity weights, best of 20 seeds, on the network pooled over all months. The resulting anchor partition gives stable colours across periods. Participation coefficient and within-community degree (Guimerà & Amaral, 2005) measure connector roles per quarter.
6. **Change over time.** Jaccard overlap of validated link sets between consecutive months and quarters, used as the "normal churn" baseline. Monthly affinity series for chosen pairs, weekly commenter counts and the share of newcomers.
7. **Validation.** (a) Unit tests: degree constraints, duplicates, empty communities, structureless data producing no false links, pure popularity producing no hub links, planted communities recovered. (b)

Equivalence with the reference `bicm` package on real data. (c) Exact Poisson-binomial check near the cut-off. (d) A stricter threshold counting only users with ≥ 2 comments in a subreddit. (e) Artifact tests that recompute every story number from the published files.

Validation results. On a 150,000-user sample of 2018 Q1, our implementation and `bicm` v3.4.0 validate the identical set of 952 links (overlap 1.000); fitted probabilities differ by at most 1.1e-08. Exact Poisson-binomial p-values for the 300 pairs nearest the threshold would add 22 links and remove 0 (2.3% of validated links), so the approximation is slightly conservative. Under the stricter threshold, 83% of the whole-window links remain (quarters: 74%, 74%, 77%, 77%), and community assignments agree at ARI 0.933. Of the 108 automated tests, 107 run in the standard environment and pass; the remaining one compares directly against the optional `bicm` package and passes when it is installed.

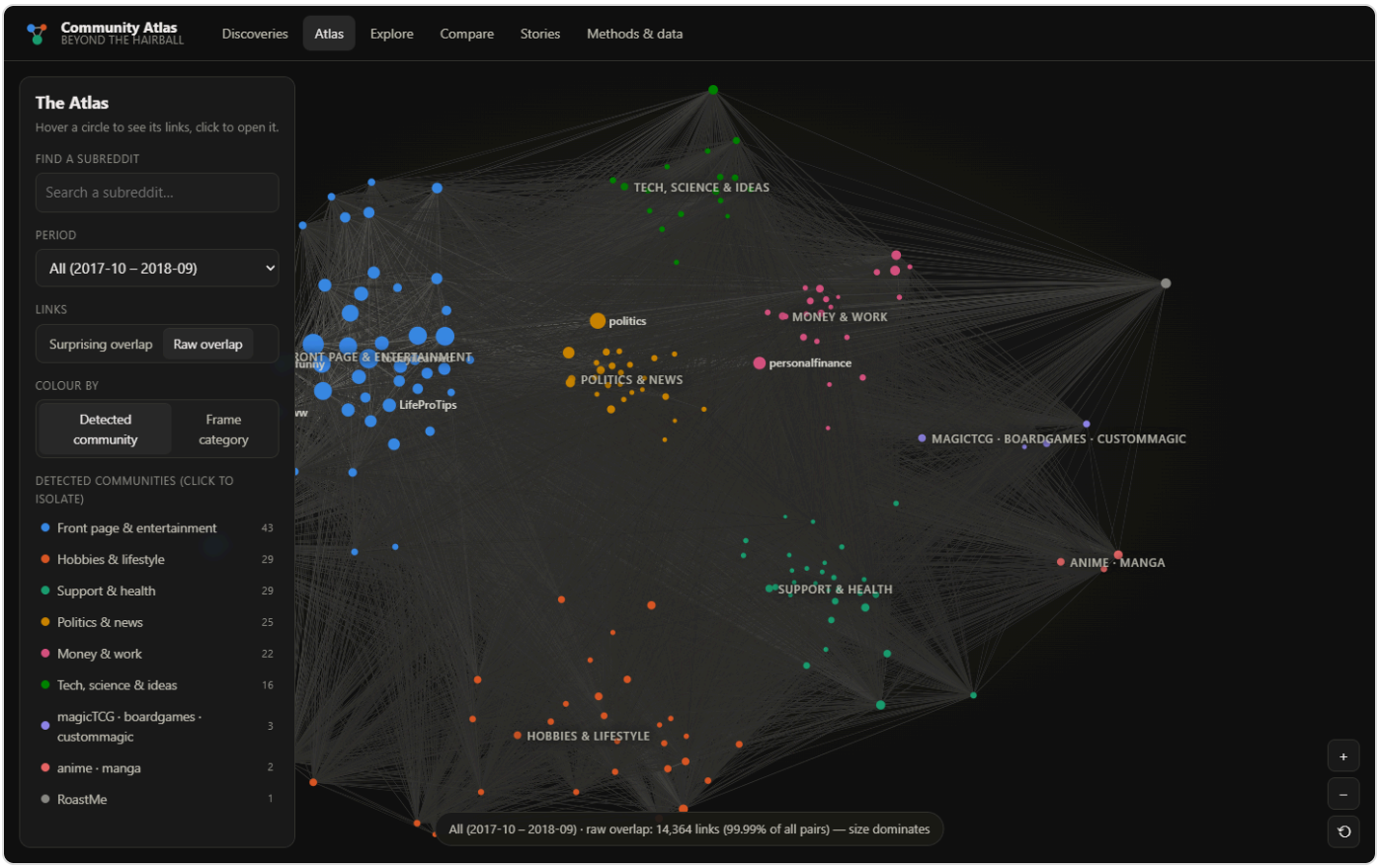
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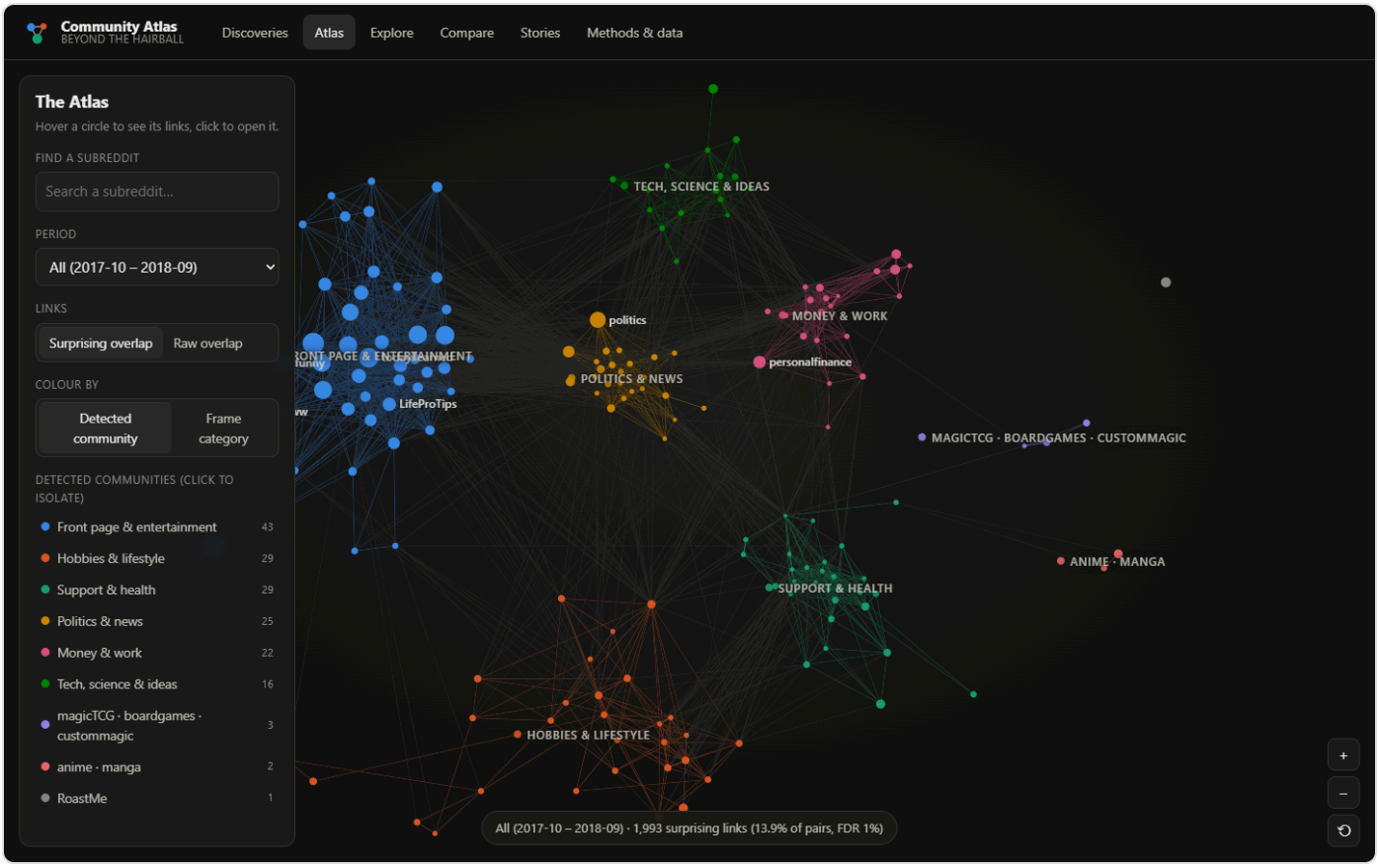
5. Results

Each result is reported as **observed → how measured → what it means → what we cannot conclude**.

5.1 What popularity hides (RQ1)

- **Observed.** Raw overlap links 99.99% of pairs over the whole window (99.91%, 99.85%, 99.93%, 99.86% per quarter). The ten communities with the largest total raw overlap are all front-page hubs (r/todayilearned, r/pics, r/AskReddit, r/news, r/worldnews, …). Among the ten closest neighbours of each topical subreddit, hubs make up **49%** by raw overlap but **4%** by affinity. Neighbours from the subreddit's own topical category rise from 36% to 66%. The validated network keeps 13.9% of pairs over the whole window, 11.1%, 11.4%, 11.5%, 11.6% per quarter and 7.2%–8.8% per month.
- **Example.** r/personalfinance's closest raw neighbours are r/news, r/LifeProTips (hub), r/AskReddit (hub), r/pics (hub), r/todayilearned (hub). Its strongest surprising overlaps are r/financialindependence (2.11×), r/RealEstate (1.94×), r/Insurance (1.78×), r/investing (1.59×), r/Frugal (1.51×). r/AskReddit, with 1,526,775 commenters, has a median affinity of 0.80× with other communities: no more than chance.
- **Meaning.** Most of what a raw shared-audience map shows is community size. Controlling for size and user activity recovers topical relationships.
- **Cannot conclude.** Affinity measures co-commenting, not friendship, conversation or influence.





5.2 Communities and connectors (RQ2)

The validated network pooled over all months has **9 communities**: weighted modularity 0.577, mean agreement between 20 Leiden seeds ARI 0.995. The communities found in each quarter agree with this anchor partition at ARI 0.887, 0.854, 0.933, 0.903. They largely follow the frame categories (ARI 0.481), with informative exceptions: r/Economics sits with politics, and media subreddits split between the front-page cluster and hobbies.

Community (auto-labelled)	Subreddits	Largest members	Category mix
Front page & entertainment	43	r/AskReddit, r/funny, r/pics, r/todayilearned, r/worldnews	front 19, media 15, tech, 4, hobbies 3, politics 2
Hobbies & lifestyle	29	r/hiphopheads, r/Fitness, r/Frugal, r/cats, r/gardening	hobbies 17, media 7, tech, 2, support 2, money 1
Support & health	29	r/relationship_advice, r/depression, r/offmychest, r/Parenting, r/Anxiety	support 23, tech, 3, media 1, politics 1, money 1
Politics & news	25	r/politics, r/PoliticalHumor, r/conspiracy, r/Libertarian, r/changemyview	politics 21, tech, 2, money 2
Money & work	22	r/personalfinance, r/Bitcoin, r/CryptoCurrency, r/wallstreetbets, r/investing	money 21, politics 1
Tech, science & ideas	16	r/buildapc, r/programming, r/sysadmin, r/learnprogramming, r/linux	tech, 14, hobbies 2
magicTCG · boardgames · custommagic	3	r/magicTCG, r/boardgames, r/custommagic	hobbies 3
anime · manga	2	r/anime, r/manga	media 2
RoastMe	1	r/RoastMe	front 1

Connectors. The participation coefficient is 0 when all of a subreddit's surprising links stay inside its own community and approaches 1 when they spread evenly across communities. Among subreddits with at least 8 links, the highest scores in every quarter belong to mid-sized topical communities, not hubs:

Subreddit	Frame category	Clusters reached	Participation (min over quarters)
r/Frugal	Money & work	5	0.65
r/Economics	Money & work	6	0.63
r/askscience	Tech, science & ideas	5	0.60
r/business	Money & work	6	0.59
r/woodworking	Hobbies & lifestyle	4	0.58
r/NoStupidQuestions	Front page & entertainment	4	0.57
r/childfree	Support & health	3	0.56
r/askphilosophy	Tech, science & ideas	4	0.55

Front-page hubs average 0.25, topical communities 0.31. Connector rankings are stable between quarters (Spearman ρ 0.87, 0.90, 0.94). *Cannot conclude:* a connector shares unusually many commenters with several clusters; that is not evidence that information flows through it.

5.3 Unexpected neighbours (RQ2)

218 links join subreddits from different frame categories, excluding hubs, and are significant in every quarter under both thresholds. The strongest:

Pair	Categories	Shared (all months)	Expected	Minimum affinity in any quarter
r/CanadaPolitics – r/PersonalFinanceCanada	Politics & news × Money & work	1,536	249	7.07×
r/UKPersonalFinance – r/ukpolitics	Money & work × Politics & news	2,015	359	6.54×
r/Economics – r/geopolitics	Money & work × Politics & news	1,393	281	5.26×
r/startups – r/webdev	Money & work × Tech, science & ideas	1,291	303	5.03×
r/askphilosophy – r/CapitalismVSocialism	Tech, science & ideas × Politics & news	320	72	4.80×
r/CapitalismVSocialism – r/Economics	Politics & news × Money & work	833	211	4.50×
r/Economics – r/PoliticalDiscussion	Money & work × Politics & news	2,257	565	4.46×
r/audioengineering – r/Guitar	Tech, science & ideas × Hobbies & lifestyle	1,639	411	4.41×

Two of the three strongest pair a country's politics forum with the same country's personal-finance forum: for these communities, nationality organises audiences more than topic. Links that also cross the detected communities include r/CanadaPolitics – r/PersonalFinanceCanada (≥ 7.07×), r/startups – r/webdev (≥ 5.03×), r/business – r/inthenews (≥ 2.96×), r/nutrition – r/selfimprovement (≥ 2.87×), r/Entrepreneur – r/webdev (≥ 2.79×), r/ChronicPain – r/crochet (≥ 2.71×). *Cannot conclude:* why audiences overlap. Shared geography, profession or demographics are all possible, and without text we cannot tell them apart.

5.4 Stability and crowds (RQ3)

Stability. Consecutive quarters share 0.83, 0.82, 0.85 of their validated links (Jaccard); consecutive months share 0.78–0.82. Change smaller than that is normal churn.

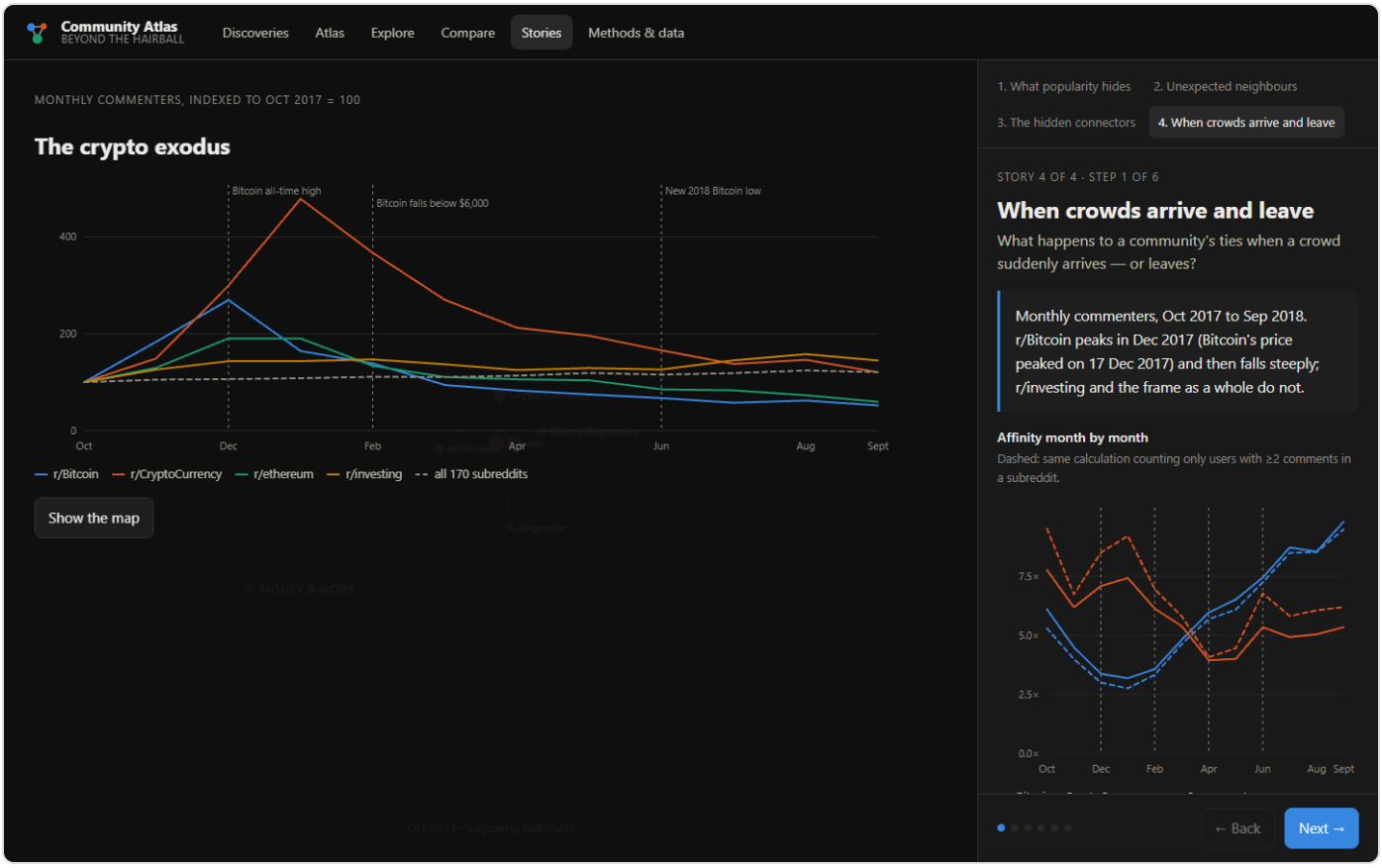
The crypto exodus. r/Bitcoin's monthly commenters peak in Dec 2017, the month Bitcoin's price peaked (17 Dec 2017). From Dec 2017 to Sep 2018, r/Bitcoin -81%, r/CryptoCurrency -60%, r/ethereum -69%, r/dogecoin -68%, r/BitcoinBeginners -88%, while r/investing changed +1% and the whole frame +14%.

Meanwhile the remaining audiences overlapped far more: r/Bitcoin–r/CryptoCurrency affinity was 6.1× in Oct 2017, fell to 3.37× at the peak as newcomers arrived, and rose to **9.8×** by Sep 2018 (9.48× under the stricter threshold).

The Infinity War crowd. *Avengers: Infinity War* opened in the US on 27 April 2018. r/marvelstudios went from a March median of 12,229 weekly commenters to 24,196 in the release week. In April, r/marvelstudios–r/Marvel affinity fell to 3.94× (from 7.42× in January 2018) and recovered to 5.34× by Sep 2018.

General tendency. Across all communities, month-to-month audience growth goes with falling affinity to their usual neighbours, and shrinkage with rising affinity, but modestly: Spearman ρ = -0.452 over 128 swings larger than 25% (ρ = -0.111 over all 1,771 month-steps).

- **Meaning.** An arriving crowd consists largely of people who comment in only one of the related communities, which dilutes surprising overlap. A departing crowd leaves behind a core that is active across them.
- **Cannot conclude.** That the price crash or the film *caused* these changes. Event dates are external context and the patterns are temporal associations.

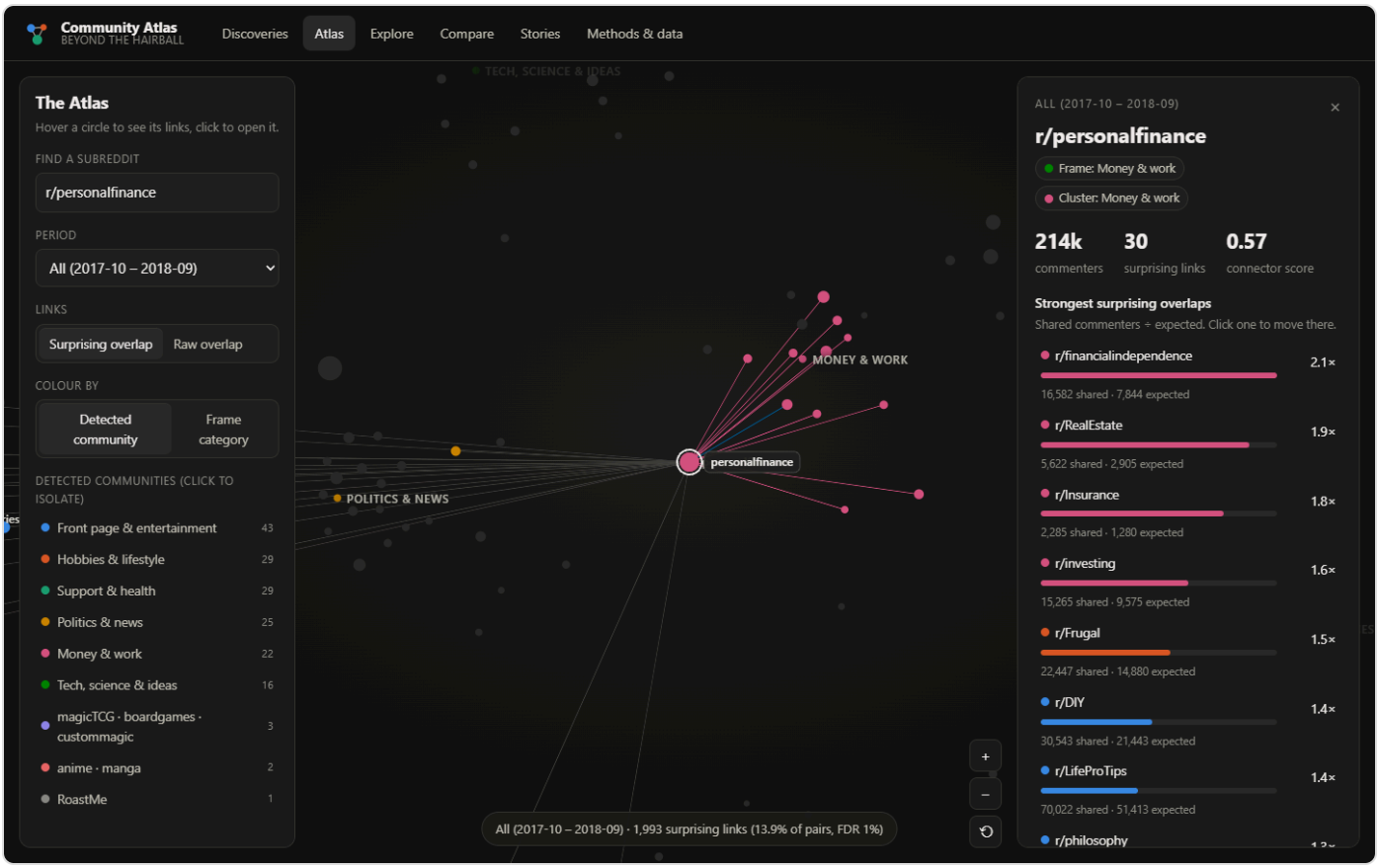


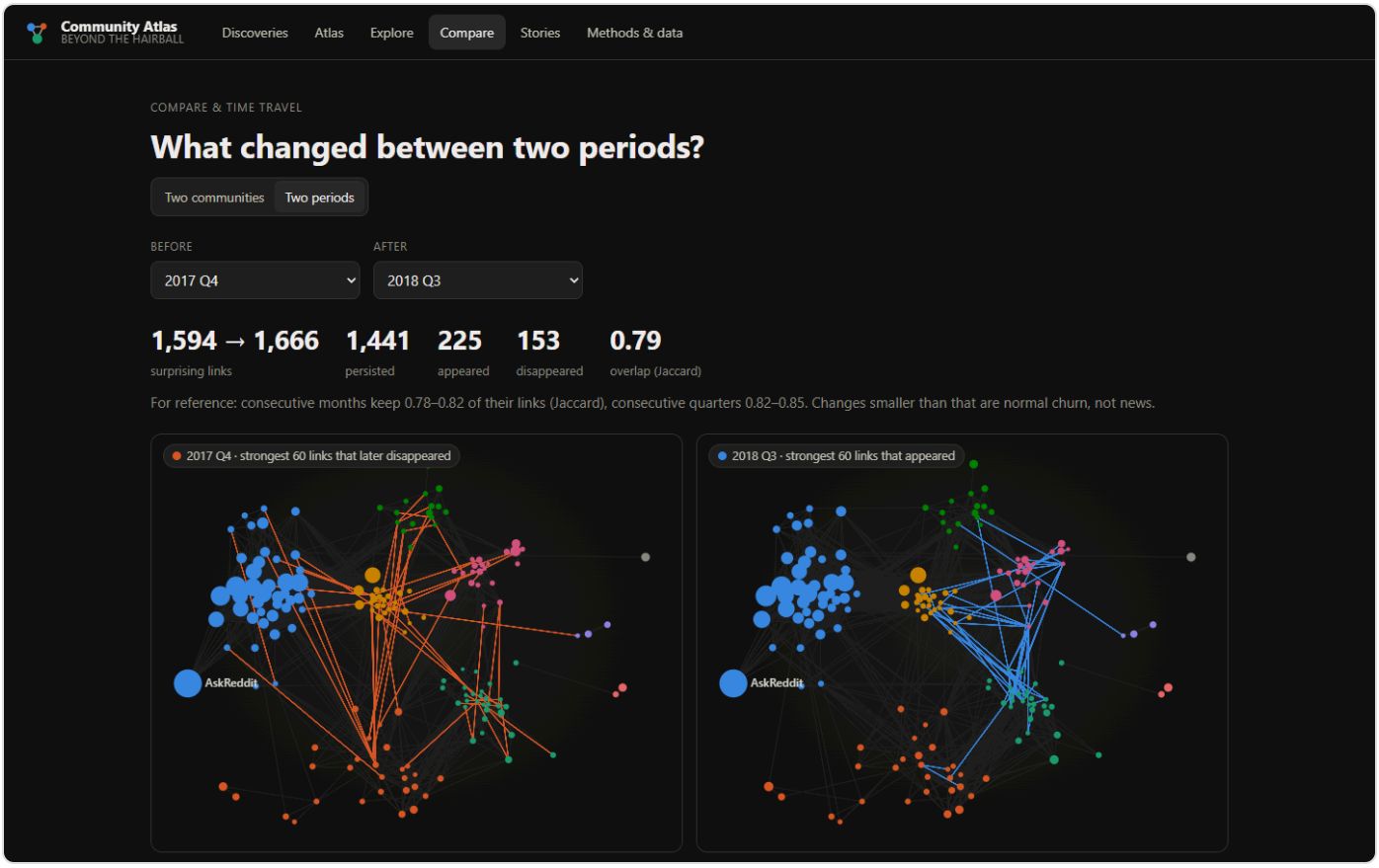
6. The application: the Community Atlas

The Atlas is a static web application (HTML/JavaScript with sigma.js, all libraries bundled) that reads only the aggregate JSON produced by `atlas/build_atlas.py`. Every view is addressable by URL, so a view can be shared or cited.

- **Discoveries:** the question, key numbers, a live raw ↔ adjusted toggle, and the four stories.
- **Atlas map:** search; period selector (whole window, quarters, months); raw ↔ adjusted links; colour by detected community or frame category; community filter; hover details; a side panel with the selected subreddit's strongest overlaps (observed vs expected) and what raw overlap would have shown.
- **Explore:** a profile for any subreddit: ranked surprising overlaps, raw-vs-adjusted neighbours, weekly commenters, monthly affinity with its strongest neighbours (significant months marked), and its connector role by quarter.
- **Compare:** two subreddits (verdict, monthly affinity, per-quarter table, shared neighbours), or two periods (side-by-side maps with appeared and disappeared links, against the churn baseline).
- **Stories:** four guided walkthroughs that drive the map, show their evidence and caveats, and end with "keep exploring from here".
- **Methods & data:** scope, cleaning, the metric, validation, project evolution, privacy, limitations and downloads (CSV and GEXF for Gephi and Tableau).

The existing Streamlit dashboard remains as the **Data & Pipeline Console** (coverage, cleaning audit, bot review).


Community Atlas
BEYOND THE HAIRBALL
Discoveries
Atlas
Explore
Compare
Stories
Methods & data
The Atlas
Hover a circle to see its links, click to open it.
FIND A SUBREDDIT
r/personalfinance
PERIOD
All (2017-10 – 2018-09)
LINKS
Surprising overlap
Raw overlap
COLOUR BY
Detected community
Frame category
DETECTED COMMUNITIES (CLICK TO ISOLATE)
Front page & entertainment 43
Hobbies & lifestyle 29
Support & health 29
Politics & news 25
Money & work 22
Tech, science & ideas 16
magicTCG · boardgames · custommagic 3
anime · manga 2
RoastMe 1
TECH, SCIENCE & IDEAS
MONEY & WORK
personalfinance
POLITICS & NEWS
All (2017-10 – 2018-09) · 1,993 surprising links (13.9% of pairs, FDR 1%)
ALL (2017-10 – 2018-09)
r/personalfinance
Frame: Money & work
Cluster: Money & work
214k
commenters
30
surprising links
0.57
connector score
Strongest surprising overlaps
Shared commenters ÷ expected. Click one to move there.
r/financialindependence 2.1×
16,582 shared · 7,844 expected
r/RealEstate 1.9×
5,622 shared · 2,905 expected
r/Insurance 1.8×
2,285 shared · 1,280 expected
r/investing 1.6×
15,265 shared · 9,575 expected
r/Frugal 1.5×
22,447 shared · 14,880 expected
r/DIY 1.4×
30,543 shared · 21,443 expected
r/LifeProTips 1.4×
70,022 shared · 51,413 expected
r/philosophy


Community Atlas
BEYOND THE HAIRBALL
Discoveries
Atlas
Explore
Compare
Stories
Methods & data
COMPARE & TIME TRAVEL
What changed between two periods?
Two communities
Two periods
BEFORE
2017 Q4
AFTER
2018 Q3
1,594 → 1,666
surprising links
1,441
persisted
225
appeared
153
disappeared
0.79
overlap (Jaccard)
For reference: consecutive months keep 0.78–0.82 of their links (Jaccard), consecutive quarters 0.82–0.85. Changes smaller than that are normal churn, not news.
2017 Q4 · strongest 60 links that later disappeared
AskReddit
2018 Q3 · strongest 60 links that appeared
AskReddit

7. How the project evolved

1. **Proposal.** Test whether bridge positions between communities persist while their occupants change, using a cosine projection, Leiden communities, participation-coefficient brokers and a turnover null model.
2. **V1 on real data** (tag `v1-baseline`). The cosine network linked 99.85% of pairs in 2018 Q1. Leiden found 4 communities, so only 6 bridge positions existed, all saturated by thousands of brokers. Pre-registered quality and sample-size gates failed, and all temporal results were labelled descriptive.
3. **Audit.** V1's reported community quality (Q = -0.0041) was *unweighted* modularity: leidenalg's `.modularity` ignores edge weights. The same partitions have weighted modularity 0.1251–0.129 against a shuffled-label maximum near 0 (`v1_modularity_audit.json`). V1's communities were real but weak; the bridge design was limited by the representation.
4. **Re-representation.** Literature on bipartite projections pointed to null-model validation (§2). On real data, BiCM-validated links were about 12% dense per quarter, stable over time, and reproduced exactly by the reference implementation.
5. **Final deliverable.** The Atlas keeps the bridge idea as the connector role and the original temporal interest in the crowd stories. The occupant-turnover hypothesis remains future work: it would need many more bridge positions than V1 could provide, and a properly matched null model.

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8. Limitations and ethics

- **Sample and window.** 170 subreddits and 12 months; overlaps with communities outside the frame are not observed. The frame's "gateway" stratum is large: front-page hubs hold most comments.
- **Commenters only.** Lurkers and voters are invisible. Comments deleted before archiving are missing, and Pushshift ingestion had outages (Gaffney & Matias, 2018), so counts are lower bounds.
- **No text.** Co-commenting is not conversation or influence; motives and topics cannot be inferred.
- **Statistical choices.** The test approximates the Poisson-binomial (slightly conservative); the FDR level and the activity threshold change details. The number of communities depends on thresholds (ARI 0.933 under the stricter one).
- **Events.** Associations, not causal effects.
- **Labels.** Frame categories come from a keyword rule and can be wrong. The network suggests r/smoking (labelled “Support & health”) is a smoked-food community: its strongest overlaps are r/Cooking, r/newsokunomoral, r/woodworking, and it peaks in Jul 2018.
- **Ethics and privacy.** Data are public. Usernames are salted and hashed at collection, and the salt is never committed. Only subreddit-level aggregates are published, which an automated test checks. Mental-health communities appear only as aggregates. Earlier commits of this repository contain a bot-review file of 200 hashed account IDs from a prototype run; it is now untracked and ignored, and purging it from the remote history is recommended.

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9. Conclusion

Raw shared-audience networks mostly measure size. Once size and user activity are accounted for, about one in seven subreddit pairs in our sample shares a surprisingly large audience. Those links form stable topical communities, reveal cross-topic neighbours (a nation's politics and personal-finance forums) and connector communities (r/Frugal, r/Economics, r/askscience), and respond to crowds arriving and leaving. The Community Atlas lets an evaluator verify each of these claims interactively. **Future work:** extend the window to 2019–2022 with the same pipeline; revisit bridge-occupant turnover with link-level bridge positions; add a reply-network layer (`parent_id` is available in the source) to separate co-presence from conversation.

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Appendix A: where each headline number comes from

Claim	Value	Artifact
Raw comments / retained	223,218,978 / 210,325,806	`data_curation/data/logs/cleaning_report.csv`
Distinct commenters	2,446,462	`atlas/web/data/meta.json` → `summary.users_total`
Raw-overlap density (all months)	99.99%	`meta.json` → `summary.cosine_density.all`
Validated density (all months / quarters)	13.9% / 11.1%, 11.4%, 11.5%, 11.6%	`meta.json` → `summary.validated_density`
Communities, weighted Q, seed ARI	9, 0.577, 0.995	`meta.json` → `summary.anchor`
Hub share of top-10 neighbours (raw / adjusted)	49% / 4%	`stories.json` → `popularity.metrics`
Robust cross-category links	218	`stories.json` → `neighbours.metrics`
Top connector participation	r/Frugal 0.65	`stories.json` → `connectors.metrics` ; `nodes.json`
r/Bitcoin–r/CryptoCurrency affinity (Oct 2017 / Dec 2017 / Sep 2018)	6.1 / 3.37 / 9.8	`stories.json` → `crowds.metrics.crypto_pair_affinity` ; recomputed by `tests/test_atlas_artifacts.py`
Quarter-to-quarter link Jaccard	0.835, 0.825, 0.852	`meta.json` → `summary.quarter_edge_jaccard`
Reference cross-check	952 = 952 links	`atlas/validation/bicm_crosscheck.json`
V1 stored vs weighted Q (2018 Q1)	-0.0041 vs 0.1251	`data_curation/data/logs/v1_modularity_audit.json`

Appendix B: reproducing everything

```
pip install -r requirements.txt
python data_curation/run_pipeline.py --start 2017 10 --end 2018 9   # fetch + clean (needs a local
python atlas/build_atlas.py                                          # all Atlas data, stories and
python -m pytest tests/                                              # 108 tests (1 needs the optic
python atlas/serve.py                                                # open the Atlas
python tools/build_docs.py --pdf                                     # this report
```

Report generated 2026-10-28 from build 2026-10-28T00:20:39+00:00 (commit 7ccac45).